



TECHNICAL NOTE

TN104

IRS COMMAND INTERFACE

REV: 2.1

REVISION HISTORY

Rev	Date	Change Description	Prepared By	Verified By	Approved By
1.0	16/01/2023	Applicable to Z175/ID09	M. Rispoli	L.	

CHANGE LOG

- Compressor: Added command to set the minimum selectable target compression;
- Config: Added command to change the compressor release behavior
- System: Added a command to test the SSR relay integrity;
- Biopsy: Added command to enable/disable the use of the Y upright position sensor;

TABLE OF CONTENT

- 1. Document Overview
- 2. IRS Overview
- 3. IRS activation
- 4. PC connection setup
- 5. IRS connection establishment
- 6. Command syntax
- 7. Group command description
 - 7.1. SYSTEM COMMANDS (system:)
 - 7.1.1. **setDATE**
 - 7.1.2. **getRevisions**
 - 7.1.3. **setShotMode**
 - 7.1.4. **setShotNodeMode**
 - 7.1.5. **setCalibKv**
 - 7.1.6. **setCalibIa**
 - 7.1.7. **setCalibTomo**
 - 7.1.8. **setOperatingMode**
 - 7.1.9. **reboot**
 - 7.1.10. **setPowerOff**
 - 7.1.11. **setUnpark**
 - 7.1.12. **setPark**
 - 7.1.13. **getSSRStat**
 - 7.2. CONFIGURATION COMMANDS (config:)
 - 7.2.1. **setAutoFilter**
 - 7.2.2. **setHsStarter**
 - 7.2.3. **setArmMotor**
 - 7.2.4. **setTrxMotor**
 - 7.2.5. **setDemoMode**
 - 7.2.6. **enableAccessory**
 - 7.2.7. **setSN**
 - 7.2.8. **setPSW**
 - 7.2.9. **enableStarterKeep**
 - 7.2.10. **setTubeTemp**
 - 7.2.11. **setLanguage**
 - 7.2.12. **resetGantry**
 - 7.2.13. **resetKvCalib**
 - 7.2.14. **sysBackup**
 - 7.2.15. **sysRestore**
 - 7.2.16. **sysFactoryReset**
 - 7.3. COLLIMATION COMMANDS (collim:)

- 7.3.1.4. **updateColliU2**

- 7.3.2. 2D Paddle Collimation calibration

- 7.3.2.1. **setCalib2D**

- 7.3.2.2. **setCalibCustom**

- 7.3.3. Collimation

- 7.3.3.1. **setManual**

- 7.3.3.3. **setAuto**

- 7.3.3.4. **setL**

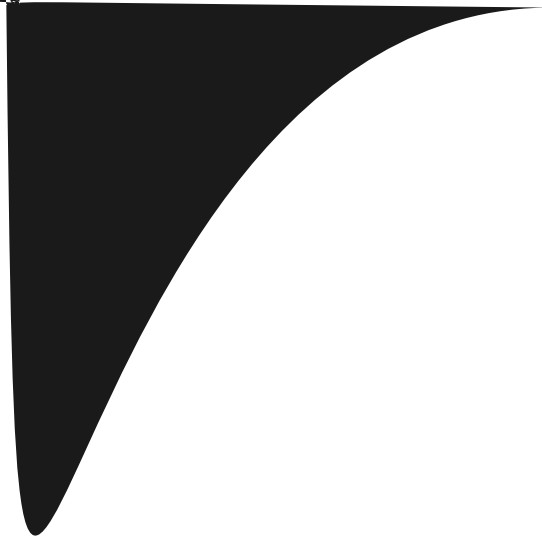
- 7.3.3.5. **setR**

- 7.3.3.6. **setF**

- 7.3.3.7. **setB**

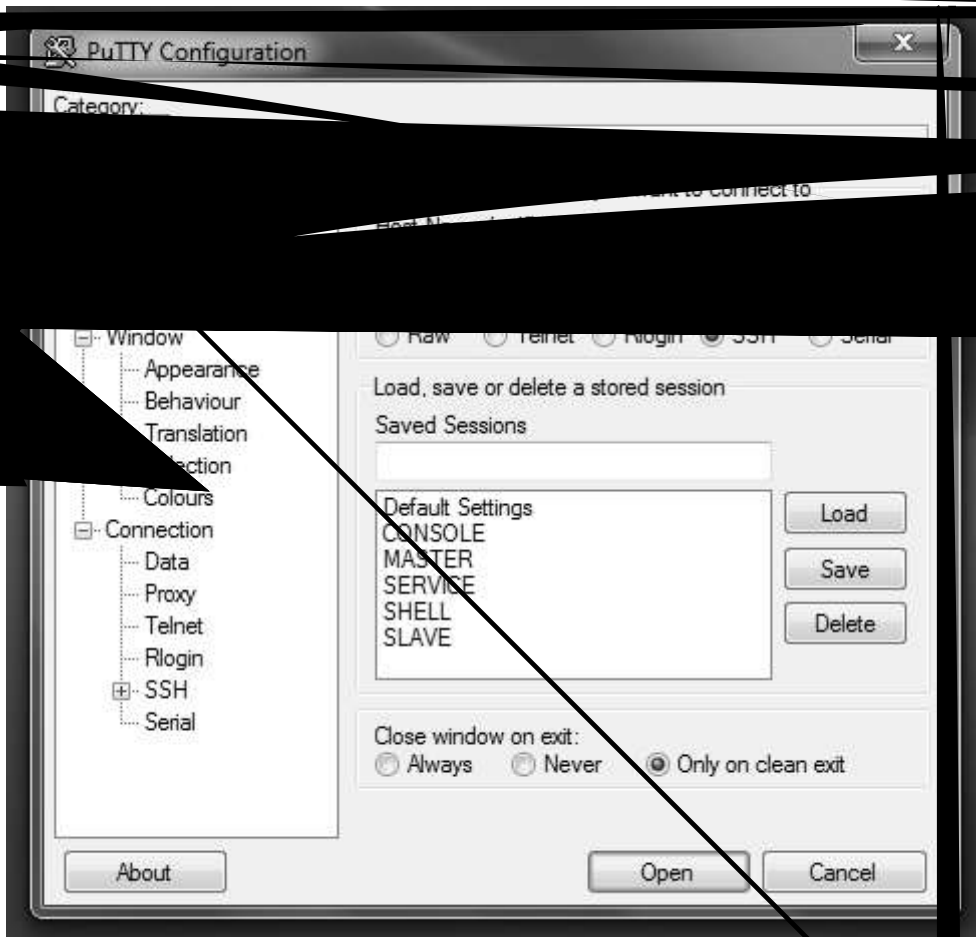
- 7.3.3.8. **setT**

- 7.3.3.9. **set**

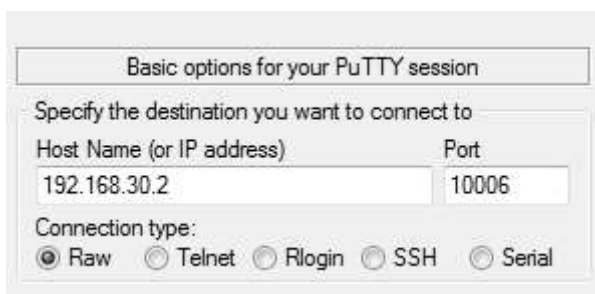


- 7.4.4. Compression Force Parameters
 - 7.4.4.1. **setWeight**
 - 7.4.4.2. **setLimitForce**
 - 7.4.4.3. **setMinTarget**
 - 7.5. MOTOR ACTIVATION COMMANDS (rotazioni:)
 - 7.5.1. Configuration File Mana
 - Config**
 - 7.5.1.2. **saveTrxConfig**
 - 7.5.1.3. **readArmConfig**
 - 7.5.1.4. **getTrxConfig**
 - 7.5.2. Gonio management
 - 7.5.2.1. **resetGonio**
 - 7.5.2.2. **getGonio**
 - 7.5.3. Activation Commands
 - 7.5.3.1. **TRX**
 - 7.5.3.2. **TRX LOOP**
 - 7.5.3.3. **ARM**
 - 7.6. POTTER COMMANDS (potter:)
 - 7.6.1. **setGrid2D**
 - 7.6.2. **setGrid3D**
 - 7.7. AWS SIMULATOR COMMANDS (aws:)
 - 7.7.1. **simulateRx**
 - 7.7.2. **getFormat**
 - 7.7.3. **pcPowerOff**
 - 7.8. NEW BIOPSY-AL COMMANDS (biopsy:)
 - 7.8.1. Inquiry commands
 - 7.8.1.1. **getAdapter**
 - 7.8.1.2. **getRevision**
 - 7.8.1.3. **getSignals**
 - 7.8.2. Calibration commands
 - 7.8.2.1. **calibXbase**
 - 7.8.2.2. **calibSh**
 - 7.8.3. Activation commands
 - 7.8.3.2. **moveLoop**
 - 7.8.3.3. **moveHome**
 - 7.8.3.4. **testBuzzer**
 - 7.8.3.5. **powerLed**
 - 7.8.4. Configuration commands
 - 7.8.4.1. **setYSensor**

- 7.9. GENERATOR COMMANDS (generator:)
 - 7.9.1. Configuration commands
 - 7.9.1.1. ***reloadTube***
 - 7.9.1.2. ***getStatistics***
 - 7.9.1.3. ***getAnodeHU***
 - 7.9.2. Activation commands
 - 7.9.2.1. ***selectFilament***
 - 7.9.2.2. ***activateStarter***
 - 7.9.3. Manual Exposure commands
-



3. Set the "Basic options.." fields as follows:



4. Press the button OPEN to start the session. If the connection is successful, you will see a

```
system:
system >
```

In the example above, the "system" sub group is entered, having access to a set of commands belonging to this group.

In order to get the command content of the group, it can be send the ? command into the group:

```
system: ?
```

NOTE: when a group is entered it is not necessary to repeat the group name w

In order to get the whole group availables, the user can type the command ? in the root of the Terminal:

Type the following commands:

```
..
?
```

The ".." command causes the Terminal enter the MAIN group (the root of all groups);

The command "?" request for the list of all available groups

7. Group command description

7.1. SYSTEM COMMANDS (system:)

7.1.1. **setDATE**

COMMAND

setDATE War Monte

are revisions in the package

COMMAND

Set the Tomo calibration Mode

NOTE: the study is automatically Opened

7.1.8. **setOperatingMode**

COMMAND

setOperatingMode

DESCRIPTION

Set the Operating mode

NOTE: the study is NOT automatically Opened

7.1.9. **reboot**

COMMAND

reboot

DESCRIPTION

The Master and Slave Terminals will reboot

7.1.10. **setPowerOff**

COMMAND

setPowerOff

DESCRIPTION

Init the System power off

NOTE: this command works only with the Closed Study

7.1.11. **setUnpark**

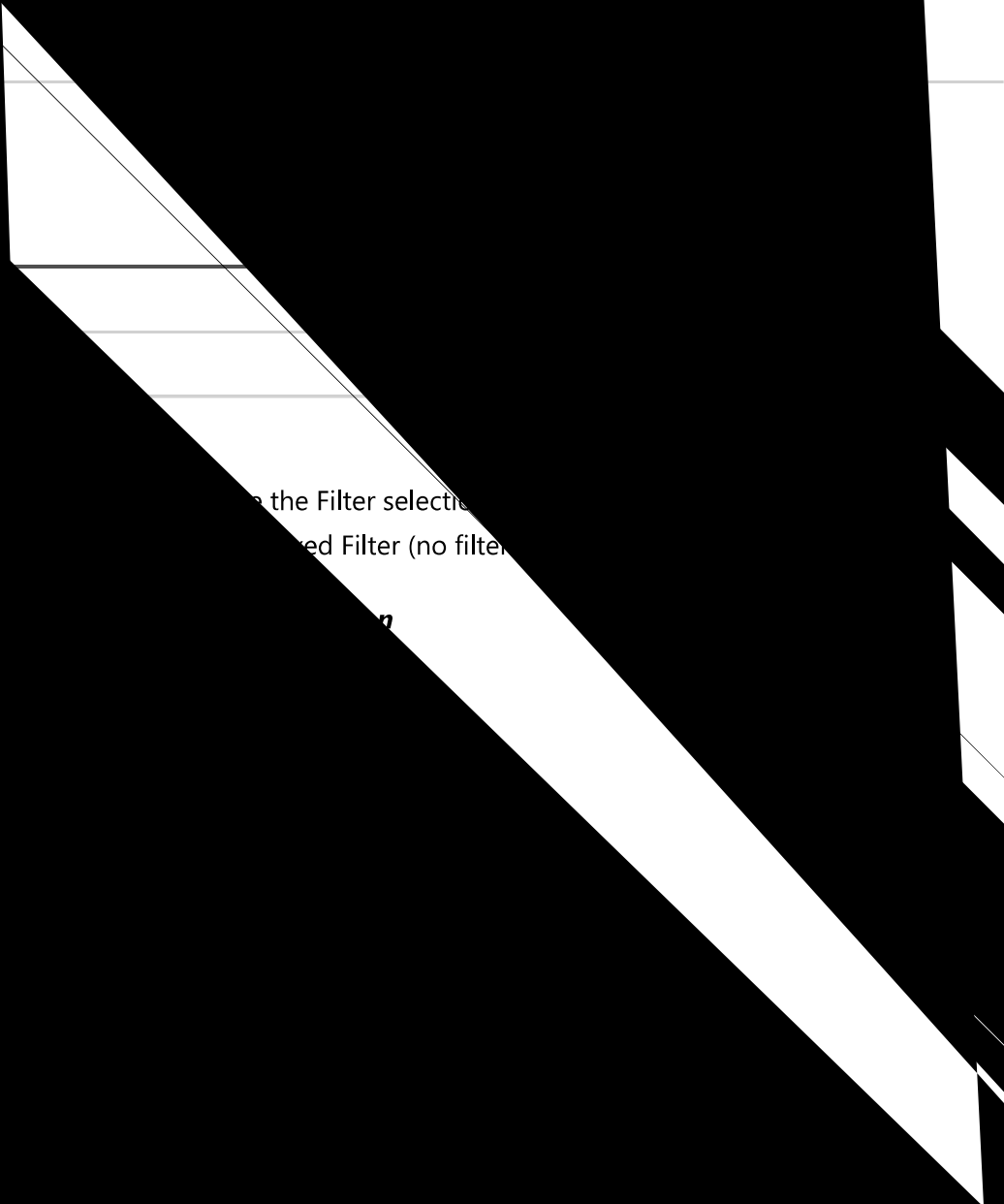
COMMAND

setUnpark

DESCRIPTION

Activate the ARM unparking procedure

7.1.12. **setPark**



COMMAND

DESCRIPTION

Set the Gantry serial number

NOTE: The `serial_number` shall be composed of only digits

Usage:

| `setSN 123456`

set the Gantry serial number to 123456

| `setSN .`

erases the serial number!

7.2.8. ***setPSW***

COMMAND

setPSW password_number

DESCRIPTION

Set the Gantry service password number

NOTE: The `password_number` shall be composed of onl

to OFF!!

7.2.10. ***setTubeTemp***

COMMAND

setTubeTemp LT HT

DESCRIPTION

Set the Tube alarm threshold(HT) and the reset alarm threshold (LT)

- HT: is the Tube temperature in (°C) to trigger the Alarm;
- LT: is the Tube temperature in (°C) to reset the Alarm;

7.2.11. **setLanguage**

COMMAND

setLanguage [ITA/ENG/SPA/FRA/POR/RUS]

DESCRIPTION

Set the GUI current language

NOTE: The Language is immediatelly changed!!

7.2.12. **resetGantry**

COMMAND

resetGantry

DESCRIPTION

Clear the Gantry System Configuration file

Use this command when the Gantry shall be reconfigured for the hardware setup:

- Motor/Manual rotation;
- HS or LS starter;
- Tilting presence;

7.2.13. **resetKvCalib**

COMMAND

resetKvCalib

DESCRIPTION

Clear the kV-sense calibration file

Use this command when the kV read back calibration shall be removed.

NOTE: In case of switch between 35kV generator to 49kV or viceversa it is recommended to clear the calibration file!!

7.2.14. **sysBackup**

COMMAND

sysBackup backup_name

DESCRIPTION

Create a backup of both Master and Slave terminals

The command creates the backup files of the Master terminal and the Slave terminal, stored in the HOME directories:

- MASTER: /home/user/master_ ***backup_name***.tar
- SLAVE: /home/user/slave_ ***backup_name***.tar

7.2.

The command doesn't update the Devices. Use the download command (see below)

7.3.1.2. **storeColliConf**

COMMAND
storeColliConf
DESCRIPTION
Stores the Collimator configuration file

7.3.1.3. **updateColliU1**

COMMAND
updateColliU1
DESCRIPTION
Updates the microcontroller U1 with the current calibration parameters related to that microcontroller

7.3.1.4. **updateColliU2**

COMMAND
updateColliU2
DESCRIPTION
Updates the microcontroller U2 with the current calibration parameters related to that microcontroller

7.3.2. 2D Paddle Collimation calibration

7.3.2.1. **setCalib2D**

COMMAND
setCalib2D Paddle Focus Left Right Front Back Trap
DESCRIPTION
Set the Paddle collimation parameters
• Paddle is one of the following paddles type: ◦ PAD24x30 ◦ PAD18x24_C ◦ PAD18x24_L ◦ PAD18x24_R ◦ PADBIOP_3D ◦ PAD9x21

	Position: 0 to 255; Collimator blade position: 0 to 255;	
	Collimator Device activation These hands allow to control the Collimator Device bypassing the Gantry	
	Collimator Device activation mode Description: Position: ASSY 01 or ASSY 02; Collimator Device: MANUALE/AUTMATICA; Collimator Filter: MANUALE/AUTOMATICA;	

When the Collimation type is AUTOMATICA then the Gantry controls the collimation.
When the Collimation type is MANUALE then the collimation is controlled manually by IRS commands.

7.3.3.2. **setManual**

COMMAND
setManual
DESCRIPTION
Set Only the collimation format in Manual mode

The Filter setting remains in Automatic mode!!

7.3.3.3. **setAuto**

COMMAND
setAuto
DESCRIPTION


command storeColliConf

7.3.5. Dynamic Filter activation

This section controls the filter position for

7.3.5.1. ***getTomoFilterConfig***

COMMAND

getTomoFilterConfig

DESCRIPTION

returns the current dynamic filter parameters

The following strings will be displayed:

- *Inseguimento filtro: abilitato* -> The dynamic filter is activated;
- *Inseguimento filtro: disabilitato* -> The dynamic filter is not active;
- *Correzione posizione: xxx* -> This is the current Adjust position value;
- *Angoli avanzamento: chg0, chg1, chg2, chg3, chg4, chg5* -> This is the current change angle vector set;

7.3.5.2. **enableTomoFilter**

COMMAND

enableTomoFilter param

DESCRIPTION

Activates/Deactivates the filter dynamic feature

- param = ON -> Activates the dynamic filter;
- param = OFF -> Deactivates the dynamic filter;

NOTE: the configuration file and the device microcontroller U2 are updated after this command!

7.3.5.3. **adjustNominalFilterPosition**

COMMAND

adjustNominalFilterPosition adj

DESCRIPTION

sets the position correction value applied to the initial filter position

- adj: signed value (positive/negative) changing the initial position of the filter.

NOTE: the configuration file and the device microcontroller U2 are updated after this command!

7.3.5.4. **setFilterChangeAngles**

COMMAND

setFilterChangeAngles chg0 chg1 chg2 chg3 chg4 chg5

DESCRIPTION

sets the array of the filter change angles, in order from negative to positive

- chg0 to chg5: signed value, in the range of [-27 to 27]

NOTE: the configuration file and the device microcontroller U2 are updated after this command!

List

DESCRIPTION

Returns the list of available compressor paddles

7.4.3.2. **setCalibPad**

COMMAND

setCalibPad offset k-force weight

DESCRIPTION

Changes the current detected paddle calibration parameters

This command allow to tune the current detected paddle parameter in order to provide a more accurate breast thickness measure.

7.4.3.3. **setThi**

no more) a phantom.

Measure the true thickness setting it
1.

Current paddle elasticity

From this calibration the User:

- Press to 150N a phantom;
- Measure the true thickness at this compression;
- Adjust the kF_val until the detected thickness equal the measured thickness.

Compression Force Parameters

Detected compression force

COMMAND

Set the minimum selectable Automatic target compression

The value can be set in the range: 30 to 200 Newton.

7.5. MOTOR ACTIVATION COMMANDS (rotazioni:)

7.5.1. Configuration File Management

7.5.1.1. *readTrxConfig*

COMMAND

readTrxConfig

DESCRIPTION

Read the TRX 

fig

DESCRIPTION

Store the ARM configuration File

7.5.1.5. *readLenzeConfig*

COMMAND

readLenzeConfig

DESCRIPTION

Read the LENZE configuration File

7.5.1.6. ***saveLenzeConfig***

COMMAND

saveLenzeConfig

DESCRIPTION

Store the LENZE configuration File

7.5.2. Incliname

COMMAND

DESCRIPTION

Activate the TRX to a target angle

The target can be of different nature:

- STOP: stops any current rotation;
- WHOME: move TRX to Wide Home position;
- NHOME: move TRX to Narrow Home position;
- IHOME: move TRX to Intermediate Home position;
- WEND: move TRX to Wide End position;
- NEND: move TRX to Narrow End position;
- IEND: move TRX to Intermediate End position;
- angle: the target is in (°). Example: TRX 10 moves TRX to 10°

7.5.3.2. **TRX LOOP**

COMMAND

TRX LOOP angle

DESCRIPTION

Activate the TRX in a loop mode

The TRX will move from +angle and -angle until a STOP command is provided.

7.5.3.3. **ARM**

COMMAND

ARM target

DESCRIPTION

Activate the ARM to a target angle

7.6. POTTER COMMANDS (potter:)

7.6.1. **setGrid2D**

COMMAND

setGrid2D ON/OFF

DESCRIPTION

Activate (ON) or Deactivate (OFF) the 2D grid (if present)

7.6.2. **setGrid3D**

COMMAND

setGrid3D ON/OFF

DESCRIPTION

Activate (ON) or Deactivate (OFF) the 3D grid (if present)

7.7. AWS SIMULATOR COMMANDS (aws:)

7.7.1. *simulateRx*

COMMAND

simulateRx frame

DESCRIPTION

Simulate the reception of an acquisition command frame

The frame shall be a valid command frame, in the format:

<ID NUM %Command param ... %>

7.7.2. *getFormat*

COMMAND

getFormat frame

DESCRIPTION

Returns information about the frame format

The command can be used to test the validity of a frame that should be simulated.

7.7.3. *pcPowerOff*

COMMAND

pcPowerOff frame

DESCRIPTION

Requests to the PC the permission to activate a Power Off sequence

The command can be used to test the validity of a frame that should be simulated.

7.8. NEW BIOPSY-AL COMMANDS (biopsy:)

The following commands refer to the Biopsy device with the lateral Approach.

inquiry commands

getAdapter

COMMAND

getAdapters

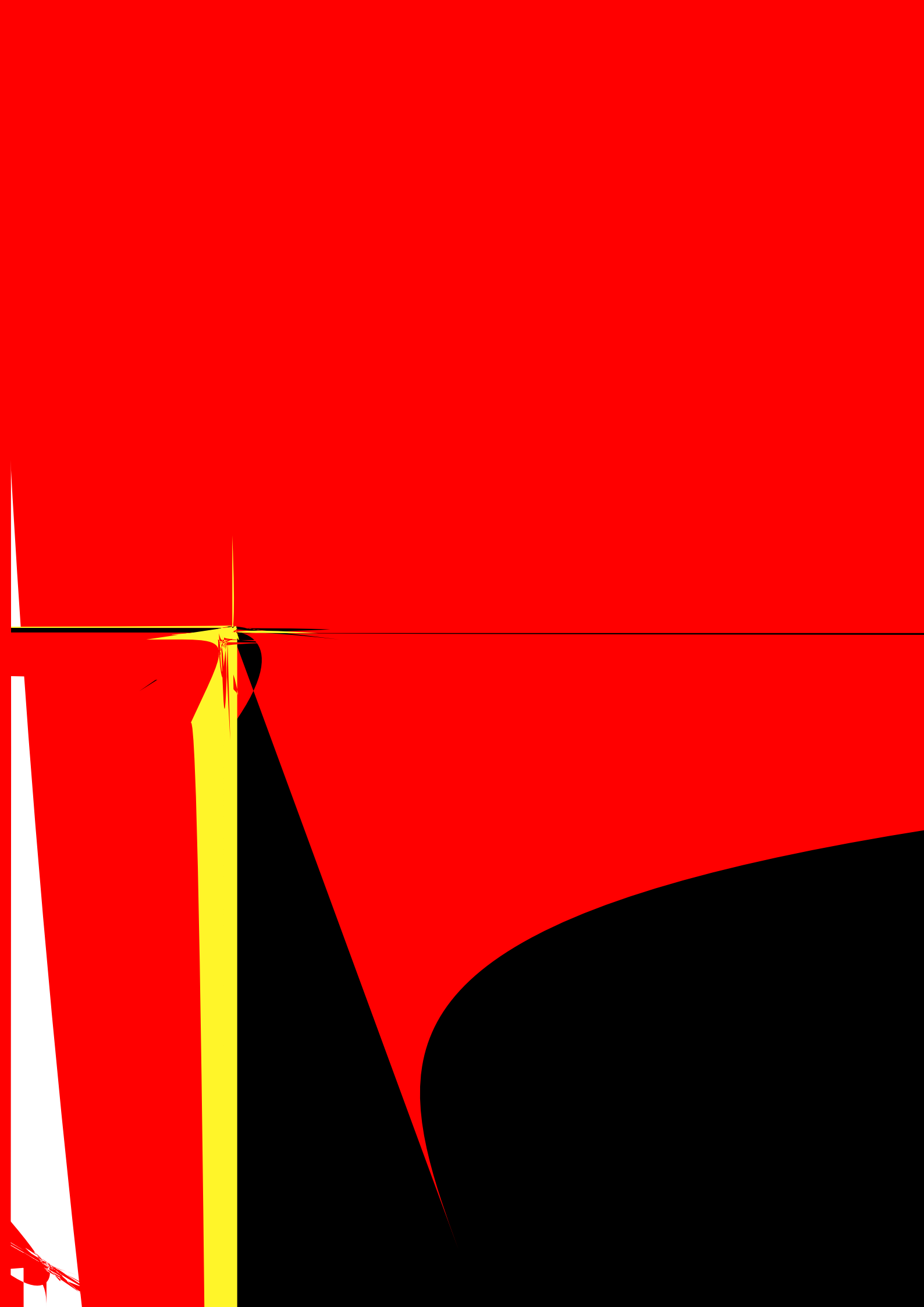
.1.3. **getSignals**

COMMAND

getSignals

DESCRIPTION

Returns the sense



7.9.3.2. **activateManualXray**

COMMAND
activateManualXray tubeType
DESCRIPTION

Activate the Manual Exposure session uploading the Tube template type

tubeType parameter:

- tubeType = "35KV": upload the TEMPLATE_35KV_XM1016T configuratin file;
- tubeType = "49KV": upload the TEMPLATE_XM1016THI configuratin file;

When the session is open the Study is Open and the Gantry is ready to expose.

The initial exposure parameters are:

- kV = 20;
- Focus = WL;
- mAs = 10;
- Idac = 2000;

7.9.3.3. **setExposure**

COMMAND
setExposure Focus kV mAs Idac
DESCRIPTION

Set the next exposure data

Focus parameter:

- Focus = "WL": will use the W large focus;
- Focus = "WS": will use the W small focus;
- Focus = "MoL": will use the Mo large focus;
- Focus = "MoS": will use the Mo small focus;

kV parameter: set 20 to 49 kV (max 35 for 35 KV template file);

mAs parameter: set in a range 1 to 10000 mAs

COMMAND

Clos

7.11.1.1. ***freeze***

COMMAND

freeze

DESCRIPTION

Stop the execution of the Mater threads

NOTE: Only Expert. Dont't use this command!!

7.11.1.2. ***run***

COMMAND

run

DESCRIPTION

Start the execution of the Mater threads

COMMAND

Send a Command frame to the **target**

the target can be:

- PCB269: this is the Compressor board;
- PCB190: this is the Generator board;
- PCB244: this is the Potter board;
- PCB249U1: this is the Collimator U1;
- PCB249U2: this is the Collimator U2;

command_code and command_param can be in hex format or decimal format;

7.11.1.8. **special**

COMMAND

special target b1 b2

DESCRIPTION

Send a Special frame to the **target**

the target can be:

- PCB269 this is the Compressor board;
- PCB190: this is the Generator board;
- PCB244: this is the Potter board;
- PCB249U1: this is the Collimator U1;
- PCB249U2: this is the Collimator U2;

b1 and b2 can be in hex format or decimal format;

7.11.2. Debug Terminals

There are two possible debugging terminal types:

- The RS232 serial terminal;
- The TcpIp terminal.

7.11.2.1. The RS232 serial terminal

The Gantry provides a DB9 connector in the back side of the cabinet providing two serial link:

- A serial link for low level debugging in the Master Terminal;
- A serial link for the low level debugging in the Slave Terminal;

After Gantry startup, those terminals remain active only for the system configuration setup time (the early steps of the Gantry startup) then they are disabled.

In order to activate the serial links the user shall use the following a IRS command:

- debug: ***enablePrint***

After the command is used both serial links will be activated until the system Power Off.

7.11.2.2. The TcpIp terinal debug

The user can open two separated Debug Terminals one for the Master activities and one for the Slave activities.

The Master terminal connection details are:

- IP: 192.168.30.2;
- PORT: 10010;
- Pro ,

- .168.30.3;
- PORT: 10010;
- Protocol Raw;

In those terminals will be conveyed both the low level activities as well the high level activities.